

DA1- DIGITAL ASSIGNMENT INSTRUCTIONS

Do a Mini Project in C, Online Hosting (WebAssembly) and PDF Documentation

What you must do

1. Develop a mini project in **C** (menu-driven)
2. Compile it to run on a browser using **WebAssembly (WASM)**
3. Host it online using **GitHub Pages**
4. Prepare a **single PDF documentation** and email it to dinakaran.m@vit.ac.in
5. **What You Will Submit – Template shared**
6. **Allowed Tools (You MAY use these)**
 - AI tools (ChatGPT / Copilot / Gemini etc.)
 - But you must **show modifications, testing, and explain your work** in PDF.
7. **Required C Concepts (Compulsory)**

Your program must include:

- **Menu-driven program** (choices using switch/case)
 - Minimum **5 user-defined functions**
 - Minimum **1 struct**
 - **File handling** (save and load data)
 - **Input validation** (range checks / wrong input handling)
8. **Online Development Options (Choose Any One)**
 - Option 1 (Recommended): Develop on your Laptop**
 - Use VS Code / CodeBlocks / Dev-C++ / GCC
 - Option 2 (If you don't want to install compiler): Online IDE**
 - Replit / OnlineGDB (only for development)
(Hosting must still be GitHub Pages using WASM.)
 9. **Hosting Option A (C → WebAssembly → GitHub Pages)**

This is mandatory in this assignment.

What is WebAssembly (WASM)?

WASM lets a browser run compiled languages (like C/C++) safely.

So your C program can run online as a webpage.

10. Step-by-Step Installation: Emscripten (Windows)

Step 1: Install Git (required)

Download and install Git:

- <https://git-scm.com/downloads>

After

Open Command Prompt and check:

git --version

Step 2: Install Python (required)

Download Python 3.x:

- <https://www.python.org/downloads/>

During

Tick: **Add Python to PATH**

Check:

python --version

Step 3: Install Emscripten SDK

Open Command Prompt (or PowerShell) and run:

installation:

installation:

1) Download emsdk

git clone https://github.com/emscripten-core/emsdk.git

2) Enter folder

cd emsdk

3) Install latest

emsdk install latest

4) Activate

emsdk activate latest

5) Set environment

emsdk_env.bat

After this, emcc will be available.

Check:

emcc -v

If it shows version details → Installed correctly.

Note: Every new terminal session needs emsdk_env.bat run once.

11. Prepare Your C Project Folder

Create a folder:

C_MiniProject

Inside it, keep:

- main.c

Example structure:

C_MiniProject/

main.c

12. Convert C Program to Web Page (WASM Compile)

Open Command Prompt inside your project folder.

Run:

emcc main.c -o index.html

This generates:

- index.html
- index.js
- index.wasm

These 3 files are your web-deployable build.

13. Important Note About Console Input/Output

A normal C program uses console input (scanf) and output (printf).

When compiled to index.html, Emscripten provides a console in the browser page.

So:

- Output appears in browser console area.
- Input can be typed in that console area.

For this assignment, that is acceptable.

(Advanced UI like buttons/forms is NOT required.)

14. Host on GitHub Pages (Free Hosting)

Step 1: Create GitHub Account

<https://github.com/>

Step 2: Create a New Repository

- Repository name example: c-mini-project-rollno
- Set it as **Public**
- Click **Create**

Step 3: Upload Files to Repository

Upload these files:

- index.html
- index.js
- index.wasm

(Optionally include main.c for reference.)

Step 4: Enable GitHub Pages

Go to your repo → **Settings** → **Pages**

Under:

Build and deployment

- Source: Deploy from branch
- Branch: main
- Folder: /root
- Click **Save**

Step 5: Get Your Live Link

GitHub will show a link like:

<https://username.github.io/repo-name/>

Open it and verify it loads.

Take screenshot of working page for documentation.

15. Documentation PDF Format (Compulsory) – Template share

16. Email Submission Instructions

Subject:

C Mini Project DA – <RegNo> – <StudentName>

Attach:

- ONE PDF only (no zip)

To: dinakaran.m@vit.ac.in

17. Evaluation Criteria

You will be scored on:

- Correctness + features
- Modular functions
- File handling
- Input validation
- Testing quality
- Successful hosting link
- Documentation quality

18. Common Issues & Fixes – Take the help of AI to address the issues and fixes.

1) emcc not recognized

Run:

- emsdk_env.bat again in that terminal.

2) GitHub Pages shows 404

Wait 2–5 minutes and refresh.

Ensure:

- repository is public
- Pages branch is main/root

3) Program not working in browser

Start with simple printf output first, compile, test.

Then add features step by step.